

Assignment:-5

Task 1:- Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

- Connect both devices to the same Wi-Fi
 - Connect your Windows laptop and your friend's Android phone to the same Wi-Fi network.
 - Make sure the phone isn't using mobile data for the test.
- Find the phone's IP address
 - On Android, open Wi-Fi settings → connected network → Network details.
 - Look for an address such as 192.168.1.105.
- Ping the phone from Windows
 - On the laptop, press Win + R, type `cmd`, and press Enter.
 - Run:

```
ping 192.168.1.105
```

Replace the example IP with the phone's actual IP.

- Interpret the result
 - Reply from 192.168.1.105... → the laptop can communicate with the phone over the network.
 - Request timed out → the devices may be isolated from each other, or the phone/network may be blocking ICMP (ping).

Task 2:- Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
C:\Users\pc>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482c:400::]
over a maximum of 30 hops:

  1    3 ms    2 ms    2 ms  2405:201:202a:a048:9ed4:a6ff:fe32:e23e
  2    *      *      *      Request timed out.
  3   15 ms   18 ms   *      2405:203:400:100:172:31:3:224
  4   12 ms   12 ms   11 ms  2405:200:801:b00::cec
  5   12 ms   11 ms   10 ms  2405:200:81a:3168:61::4
  6    *      *      *      Request timed out.
  7   11 ms   12 ms   10 ms  2405:200:801:200::8808
  8    *      *      *      Request timed out.
  9   19 ms   19 ms   19 ms  2001:4860:1:1::1b8e
 10   18 ms   18 ms   18 ms  2001:4860:1:1::1b8e
 11   18 ms   18 ms   18 ms  2001:4860:482c:400::

Trace complete.
```

Task 3:- disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for [IP address]:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Task 4:-Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
C:\Users\pc>arp -a
```

Interface: 192.168.135.1 --- 0x4

Internet Address	Physical Address	Type
192.168.135.254	00-50-56-e5-3b-d9	dynamic
192.168.135.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.29.49 --- 0x7

Internet Address	Physical Address	Type
192.168.29.1	9c-d4-a6-32-e2-3e	dynamic
192.168.29.66	94-e2-3c-dc-a4-79	dynamic
192.168.29.213	f8-54-f6-b4-ff-5d	dynamic
192.168.29.217	a8-59-5f-6e-41-0f	dynamic
192.168.29.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.155.1 --- 0x8

Internet Address	Physical Address	Type
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192.168.155.254	00-50-56-f4-e7-3b	dynamic
192.168.155.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.56.1 --- 0xa

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

Interface: 172.28.0.1 --- 0x27

Internet Address	Physical Address	Type
172.28.15.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Task 5:- Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Include screenshots or command outputs if possible.

Wi-Fi Network Testing Report

Introduction

The aim of this activity was to connect two devices to the same Wi-Fi network and check if they could communicate with each other.

1. Setting Up the Network

First, I connected my laptop and my friend's Android phone to the same Wi-Fi network. I checked the IP address of the phone from its Wi-Fi settings.

The phone's IP address was: [write IP address here]

2. Testing with Ping

I opened Command Prompt on my Windows laptop and typed:

```
ping [phone IP address]
```

The laptop received replies from the phone.

Example:

```
Reply from 192.168.1.105: bytes=32 time=5ms
Reply from 192.168.1.105: bytes=32 time=4ms
```

This means that the laptop and phone were connected and could communicate with each other.

Screenshot:

[Add a screenshot of the ping result here.]

3. Testing with Traceroute

Next, I checked the path from my laptop to YouTube. I typed:

```
tracert www.youtube.com
```

The command showed different hops between my laptop and YouTube.

The total number of hops was: [write number of hops here]

Screenshot:

[Add a screenshot of the tracert result here.]

Some lines showed `Request timed out`. This can happen because some network devices do not reply to traceroute requests.

4. Disconnecting the Phone

After that, I disconnected the phone from Wi-Fi and tried the ping command again:

```
ping [phone IP address]
```

This time, I got:

```
Request timed out.
```

This happened because the phone was no longer connected to the Wi-Fi network.

Conclusion

From this activity, I learned how devices communicate on a Wi-Fi network. The ping command helped me check the connection between the laptop and phone. The tracert command showed the path to YouTube. When the phone was disconnected, the ping failed and showed a timeout message.